PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : SURFACE CHEMISTRY (Question Bank # 11)

SECTION-I (I) ONLY ONE OPTION CORRECT TYPE

1. (B)

Sol. $A = 1.25 \times 10^8\ sec^{-1}$

$E_a = 16\ Kcal = 16.000\ Ca.$

$T = 400\ K$

$$K = Ae^{\frac{-E_a}{RT}}$$

$$= 1.25 \times 10^8\ sec^{-1}.\ e^{-\frac{16000\,cal}{2cal \times 400}}$$

$$= 1.25 \times 10^8\ sec^{-1}\ .\ e^{-20}$$

$$= \frac{1.25\times10^8}{e^{20}} sec^{-1}$$

$$= \frac{1.25\times10^8}{5\times10^8} sec^{-1} \qquad (e^{20} = 5 \times 10^8 \text{ given})$$

$$= 4\ sec^{-1}$$

$$\text{Average time } t_{avg} = \frac{1}{k} = \frac{1}{4\,sec^{-1}} = 4\,sec$$

2. (B)

Sol. Smoke precipitator work on principle of neutraligation of charge on collids.

Smoke is a collidal solution of solid charged particles, when passed over oppositively charged plate of smoke precipitator, collidal charge particles neutralies.

3. (B)

Sol. Adsorption of H_2(gas) over transition metal and forming metal hydrids is an example of chemical adsorption ans So

(A) Chemical adsorption $\propto$ surface are of metal

(B) Enthalpy change involved in rangs of 80-200 kJ/mole

(C) Chemical adsorption increases with increase in temperature upto cartain limit.

(D) Chemical adsorption will be irreversible in nature

So (B) is incorrect

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4. (A)

Sol. (A) $AuCl_3(s) + HCHO_{(l)} + H_2O_{(l)} \longrightarrow \underbrace{Au_{(s)} + HClCO + HCl_{(\ell)}}_{\text{Collidal solution (solid sol.)}}$

(B) $CO_2(g) + Mg(s) \rightarrow \underbrace{MgO(s) + C(s)}_{\text{Not collidal sol.}}$

(C) $RCOOR'_{(l)} + H_2O_{(l)} \xrightarrow{HCl_{(\ell)}} \underbrace{RCOOH(l) + R'OH(l)}_{\text{not collidal}}$

(D) $NH_3(g) + O_2(g) \longrightarrow \underbrace{NO(g) + H_2O(l)}_{\text{not collidal}}$

5. (D)

Sol. (A) Catalyst increases rate while inhibitor decreases rate

(B) Promotor work in present of catalyst

(C) In presence of metal ions like Mn^{3+},Cr^{3+} etc. activity of enzymes increases

(D) Physical adsorption decreases on increasing temperature of heat released, temperature increases.

So (D) is not correct.

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6. (C)

Sol. Crystallisation, corrosion, heterogenous catalysis occurs at interface of phases.

Homogenous catalysis occurs in same phases.

So correct answer is (C)

7. (A)

Sol. For +ve sol, it required –ve ions for coaguation.

Coagulation $\propto$ magnitude of –ve charge for +ve sol

So $[Fe(CN)_6]^{4-}$ is best for coagulation among all.

So (A) is correct option

8. (B)

Sol. Adsorption of methylene blue from aqous solution on activated charcoal is example of physical adsorption

So no. bond formation take place and no requirment of activatoin energy. It is reversible in nature and enthalpy decreases. On increasing temperature physical adsorption decreases.

So (B) is correct option.

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9. (A)

Sol. Concentration of collidal sol = 1.2×10^{-3} M

$= 1.2 \times 10^{-3}$ mole/lit

$= 1.2 \times 10^{-3} \frac{\text{mole}}{10^{-3}\,\text{m}^3}$ $\quad (1 \text{ lit} = 10^{-3}\,\text{m}^3)$

$= 1.2$ mole/m^3

So no. of moles of soap particles in 1 mm

$$= \frac{1.2\,\text{mol}}{\text{m}^3} \times 1\,\text{mm}$$

$$= \frac{1.2\,\text{mole} \times 10^{-9}}{\text{m}^3}\,\text{m}^3$$

$= 1.2 \times 10^{-9}$ mole

no. of soap particles = $1.2 \times 10^{-9} \times 6 \times 10^{23}$

So no. of soap particles per miccells

$$= \frac{\text{no. of total particles}}{\text{no. of miccells}}$$

$$= \frac{1.2 \times 10^{-9} \times 6 \times 10^{23}}{2.4 \times 10^{13}} = 30$$

So (A) is correct options.

10. (A)

Sol. Promotor - inhance (increase) the activity of catalyst

Poison - decrease the activity of catalyst

So option (A) is correct.

SECTION-II (II) ONLY MORE OPTION CORRECT TYPE

11. (A,B,C,D)

Sol. Above the C.M.C. the loss of entropy de to assembly of the surfactant molecles is less than the gain in entropy by setting free the water molecules that were traped in the solvation shell of th surfactant mononeocs.

Hydrophobic part oriented with in cluster and Hydrophilic part exposed to water to form CMC. CMC formation are dynamics. Some surfactant leave and some other enter to micells and no change in surface tension take place. So all ABCD correct

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12. (A,B,D)

Sol. Collidal solution can be parified by

Dialysis, electrodialysis, ultrafiltration.

So answer is A,B,D

13. (B,C,D)

Sol. Lyophilic sol-Gelatin sol

Lyophobic sol- Silver sol, sulphur sol, As_2S_3 sol

So ans is B,C,D

14. (A,B)

Sol. Colloidal gold can be obtained by

Bredig's arc method and reduction of $AuCl_3$

So, A, B are correct option

15. (B,D)

Sol. (A) **Emlsion:** liquid in liquid so dispersed phase and dispersion medium both should be in liquid state

(B) In adsorption both enthalpy and entropy decreases

(C) Brownian motion of collidal particles depends upon size

(D) Adsorption $\propto$ 'a' (Vander wall gas constant) $\propto T_C$

So B,D correct

16. (A,D)

Sol. Lyophobic colloidal particles are charged particles this charge genrated due to preferential adsorption of ion on their surface from solution so that a potential difference genrated in & between fixed layer and diffused layer of opposite charge.

And these like charges on each collidal particles are cause the stability of colloidal particles

So A, D are correct option.

17. (A,D)

Sol. Tyndal effect is due scattering of light with colliodal particles so repractive index of dispersed phase and-dispersion medium must differ greatly and wavelength of incoming light is not much greater than collidal particles for better scattering of light

So A,D option are correct.

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18. (A,B,C)

Sol. (A) Sol ⇒ solid in liquid

(B) Gel ⇒ liquid in solid

(C) Emulsion ⇒ liquid in liquid

(D) Aerosol ⇒ solid in gas

So A,B,C are correct option.

19. (A,B,C,D)

Sol. Kraft temperature (T_k) : That minimum temperature above which micelle formation take place

emulsion is liquid in liquid type of collidal is milk (O/W) emulsfying agent like protein and heavy metals are generaly used for stabilized emulsion

So, A,B,C,D are correct option.

20. (A,C,D)

Sol. Peptization of AgI with $AgNO_3$ form AgI-sol-In which collidal particles are positively charged i.e. AgI/Ag^+. So dispersed phase (colloidal partical, AgI/Ag^+) move towards cathode and Na_3PO_4 is more effective than $AlCl_3$ for coagulation. Persistant dialysis of sol also cause for coagulation.

So A,C,D are correct options.

21. (A,B)

Sol. From the given table we can analyse ΔT_b for solution

$$\Delta T = i.m\, K_b$$

$$\Delta T_b \propto i.m.$$

from given observation in table $\Delta T_b \propto m$ upto 0.05 mole

But of 0.06 M ΔT_b charge from pressure

linear change means 'i' factor changed after 0.05 M and before 0.06 M

So CMC must come before 0.06 M and solution becomes colloidal so at 0.07 M solution must be heterogenous also at 0.4 M. We dont know 'i' factor at 0.08 M so cannot find 'T'

So A,B correct.

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(iii) PARAGRAPH TYPE

Paragraph for question no. 22 & 33

22. (B)

Sol. $\frac{x}{m} = K.P^{\frac{1}{n}}$

$$\log\left(\frac{x}{m}\right) = \log K + \frac{1}{n}\log(P)$$

$$\frac{1}{n} = \tan\theta$$

$$\frac{1}{n} = 0.25$$

$$\boxed{n = 4}$$

At log P axis : $\log\left(\frac{x}{m}\right) = 0$

$$0 = \log K + \frac{1}{n}\log P$$

$$0 = \log K + \frac{1}{4}(-4)$$

$$\boxed{K = 10}$$

So (B) option is correct.

23. (D)

Sol. $\frac{x}{m} = KP^{\frac{1}{n}}$

$$\frac{x}{10gm} = 10.(16)^{\frac{1}{n}} = 10\times(2^4)^{\frac{1}{4}}$$

$$\frac{x}{10gm} = 10\times 2$$

$x = 200$ gm

So Ans. (D) is correct.

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SECTION-II (INTEGER SECOND DECIMAL)

1. (0.3)

Sol. **Gold number:** amount of lyophilic sol added (in miligram) into

10 ml of standered gold sol in presence of |m| of 10% NaCl solution to stop coagulation.

So G.N. = 0.030 mean

In 10 ml gold sol 0.030 miligram haemoglobin added

Now in 100 ml gold sol $\frac{0.030\times100}{10}$ = 0.30 gm haemoglobin added

Ans. 0.30

2. (6.00)

Sol. Collidal prepared by KI to $AgNO_3$ solution form +ve sol of collidal particle AgI/Ag^+

So B^- ion is causing for flocculation from electrolyte AB

Flocculation value (B^-) = 100 m mole/lit

So 1 lit solution required = 100 m mole

$= (100 \times 10^{-3}) \times 60$ gm

= 6.00 gm

3. (5.00)

Sol. (i) At surface, surface particals try to ballance force on it so adsorption take place so net de crease in residual force

(ii) $\Delta H_{adsorption} < 0$

(iii) Physisorption can be multilayered

(iv) Magneson adsorb on surface of $Mg(OH)_2$ ppt

(v) Coconut charcoal adsorb noble gases at different temperature

So all statements are true

4. (5.00)

Sol. Example of coagution are

Delta formation, precipitation of coal ash, destruction of lyophobic solution, Artificial rain, claning of water with alums.

So total 5 process among given is coagulation

5. (9.00)

$Fe_2O_3 \cdot xH_2O$ sol, basic dye are +ve sol rest are – ve sol.

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